FATURA ANALIZ

Düzce University
Faculty of Engineering

PRESENTATION 03

Design & *Development*

The system architecture, database, REST API, the Gemini microservice, and the Flutter screens — how FaturaAnaliz is built.

Student
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Supervisor
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Stack
Flutter · Spring Boot · Python

SYSTEM ARCHITECTURE

A layered client–server system, four components.

CLIENT

Flutter App

Renders screens, charts & local state; talks to the API over HTTP/JSON.

BACKEND

Spring Boot

Business logic, auth & orchestration between client and AI.

AI SERVICE

Python · Flask

Three analysis modes → Gemini 2.5 Flash API.

STORAGE

PostgreSQL

Per-user accounts, expenses & budget limits.

> Every component sits behind a clear boundary; all communication is HTTP with structured JSON.

SPRING BOOT — LAYERED BACKEND

LAYER 1

Controller

REST endpoints — routing, request/response (de)serialization only. No business logic.

LAYER 2

Service

Core logic — registration, auth, expense CRUD, aggregation, AI invocation.

LAYER 3

Repository

Spring Data JPA interfaces — typed queries per entity to PostgreSQL.

LAYER 4

Security

Spring Security — a JWT filter validates every request before it reaches a controller.

DATABASE DESIGN — POSTGRESQL

Three tables, per-user isolation.

Foreign keys tie every expense and budget limit back to its owner — enforced at both the service and database layers.

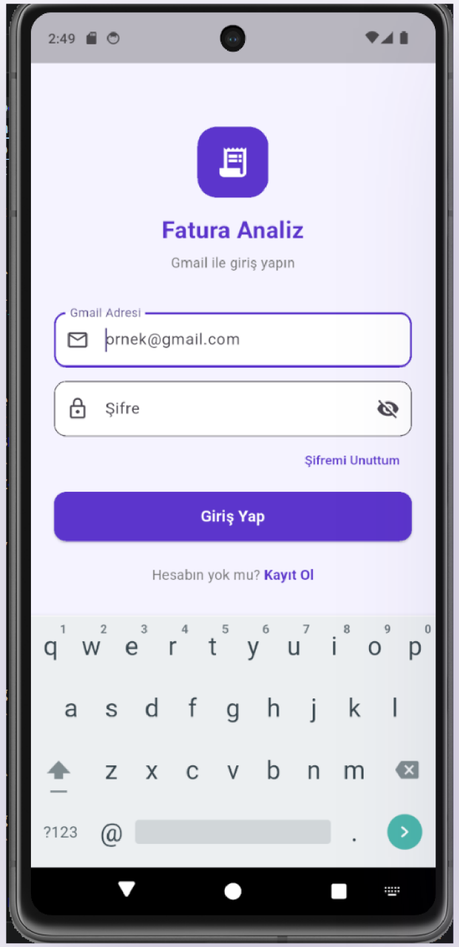
Table	Key fields
users	id · full_name · email · password (BCrypt) · is_verified · verification_token · reset_code · reset_code_expiry
expenses	id · amount · description · category · created_at · user_id (FK)
budget_limits	id · user_id (FK) · category · daily_limit · weekly_limit · monthly_limit

RESTFUL API ENDPOINTS

Auth	Endpoint
POST	/api/auth/register
GET	/api/auth/verify-email
POST	/api/auth/login
POST	/api/auth/forgot-password
POST	/api/auth/reset-password

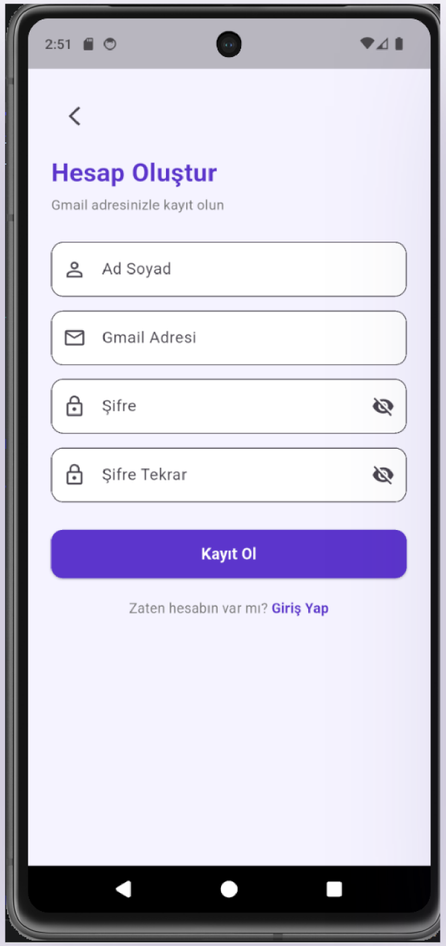
Data & AI	Endpoint
GET/POST	/api/expenses · /{id} · /stats
POST	/api/import/photo
POST	/api/import/file
GET/POST	/api/budget
GET	/api/report/monthly

> Everything outside `/api/auth/**` requires a valid Bearer JWT.



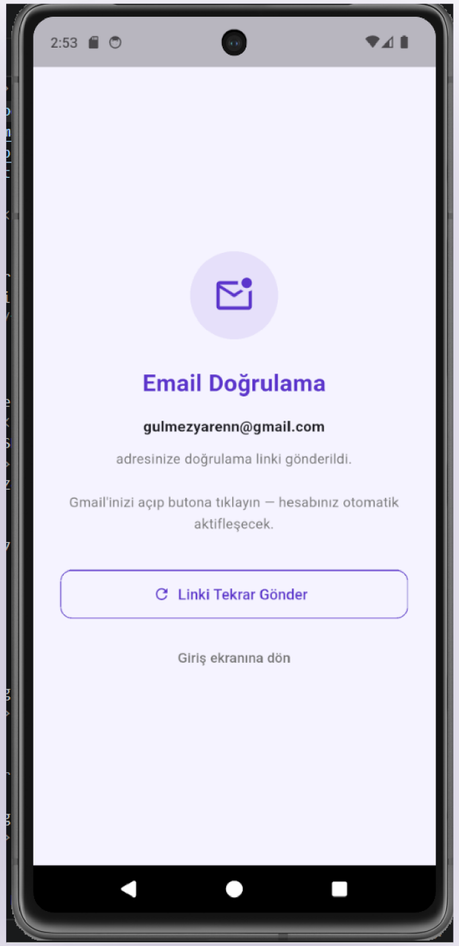
Login

Gmail + password, client-side validation



Register

Name, Gmail, password & confirmation



Verification gate

Login blocked until email is confirmed

AUTHENTICATION & SECURITY

BCrypt

Passwords stored only as salted BCrypt hashes.

JWT

Stateless tokens carry user id & email; checked every request.

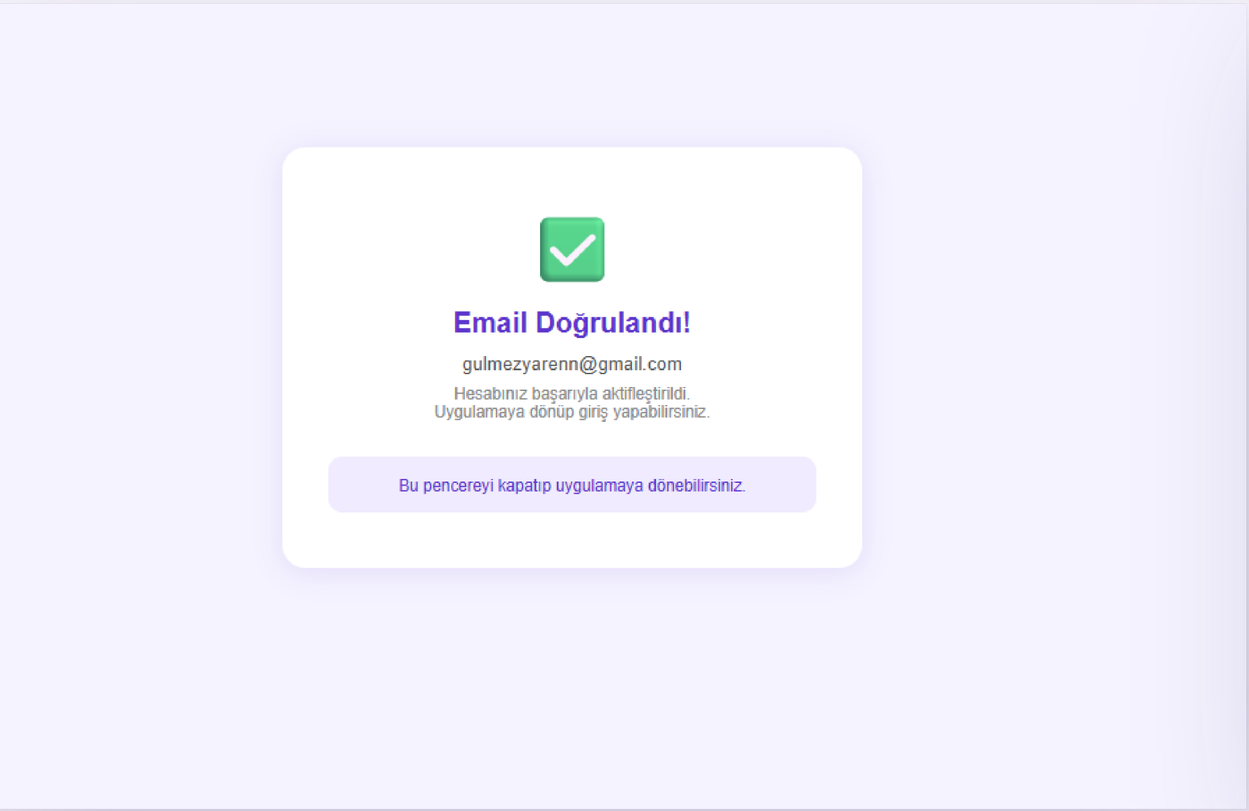
Email verify

UUID activation link, valid 24 hours, before login.

OTP reset

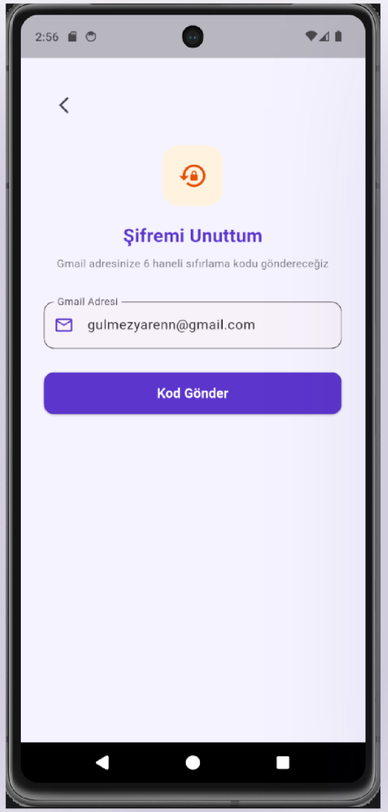
6-digit code, expires after 15 minutes.

EMAIL VERIFICATION & PASSWORD RESET



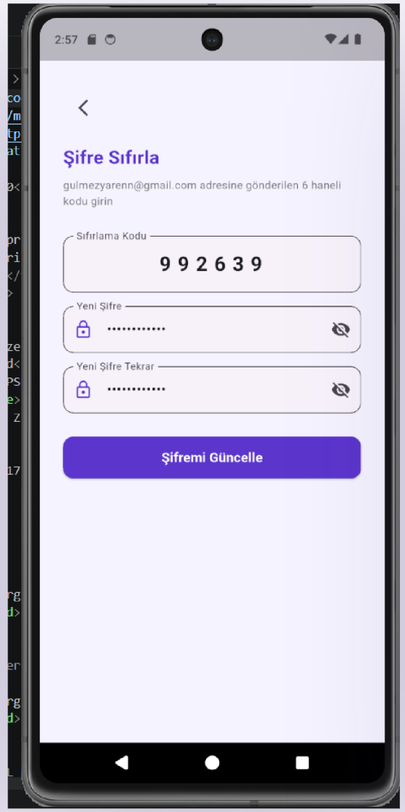
Verified

Web confirmation page



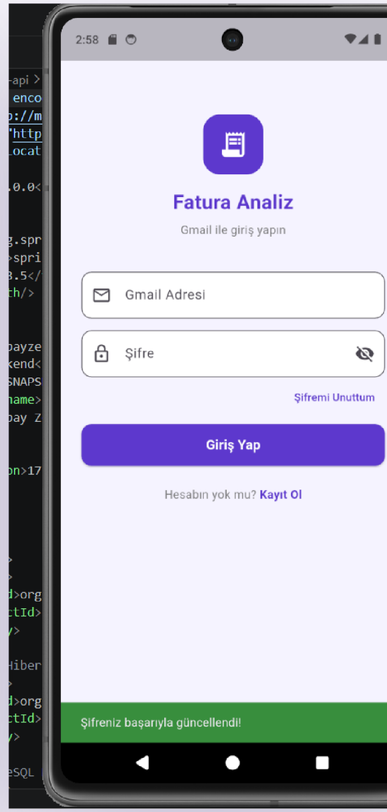
Forgot password

Request a 6-digit code



Reset

Enter OTP + new password



Success

Back to login

AI MICROSERVICE — THREE MODES

One Flask service, three Gemini prompts.

Invoice image

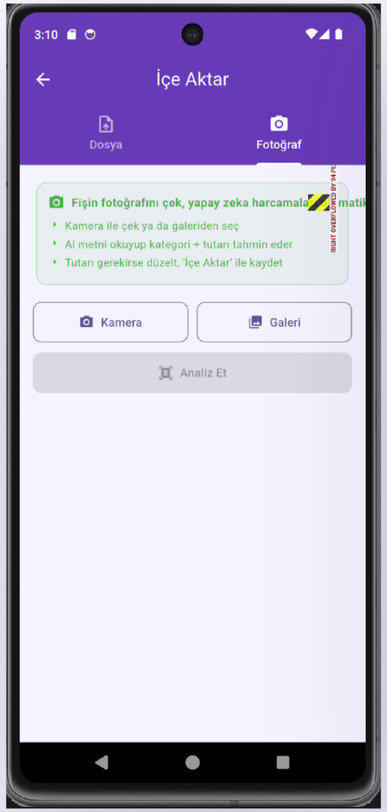
Base64 image + structured prompt → extract total & predict category in one call.

CSV / XLSX batch

pandas reads each row; description text classified into one of ten categories.

Statistical analysis

Category totals & anomalies → natural-language commentary, risk & savings.



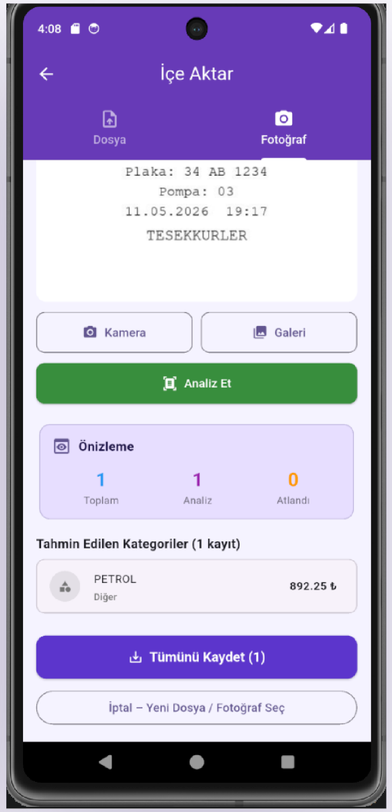
1 · Capture

Camera or gallery



2 · Analyze

Submit to Gemini 2.5 Flash



3 · Review & save

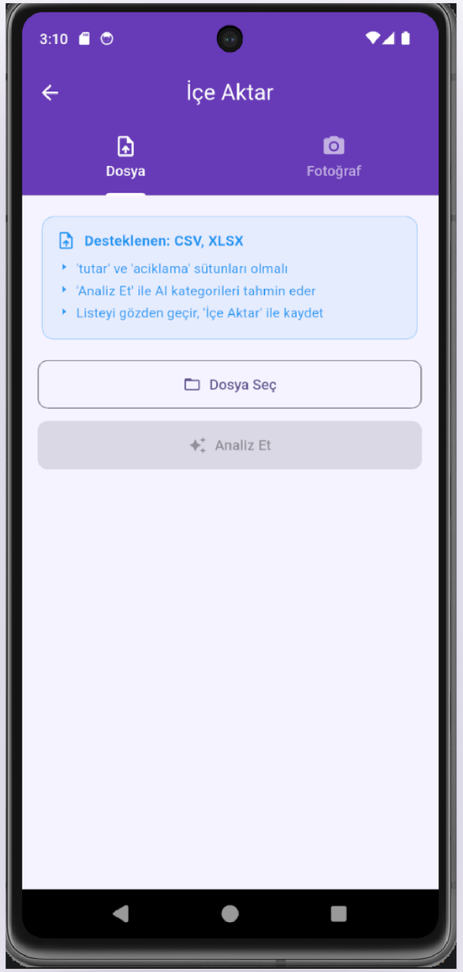
892.25 ₺ → Diğer

CSV / XLSX BATCH IMPORT

Bulk history, auto-categorized.

The file needs only `tutar` and `aciklama` columns. Each description is classified by Gemini, and the predicted records are reviewed before a one-tap bulk save.

CSV · XLSX · pandas · batch persist

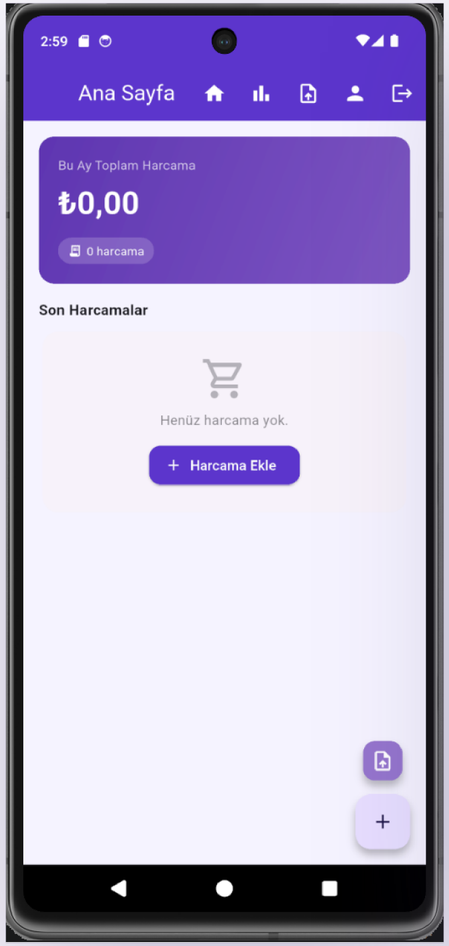


İçe Aktar — Dosya tab

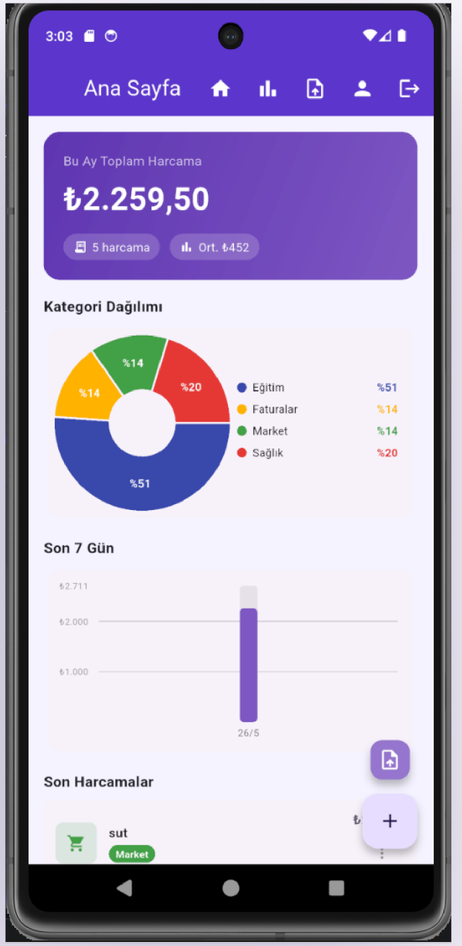
MOBILE — HOME DASHBOARD

Everything at a glance.

A monthly-total summary card, a category donut, a seven-day trend bar chart, and the recent-expense list — with two entry points: manual add and AI import.

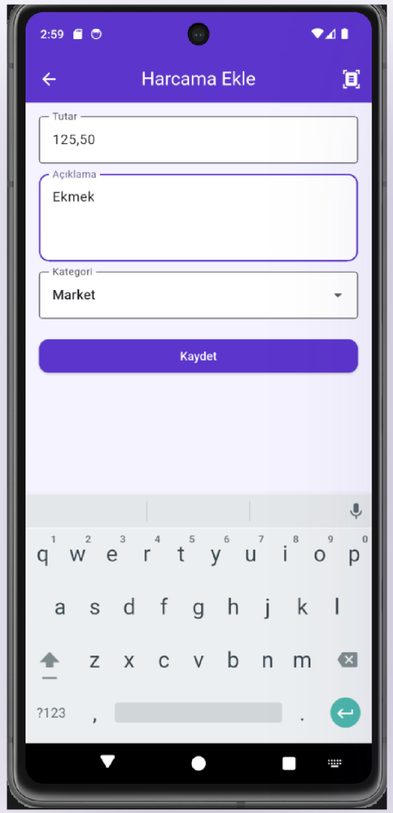


Empty state



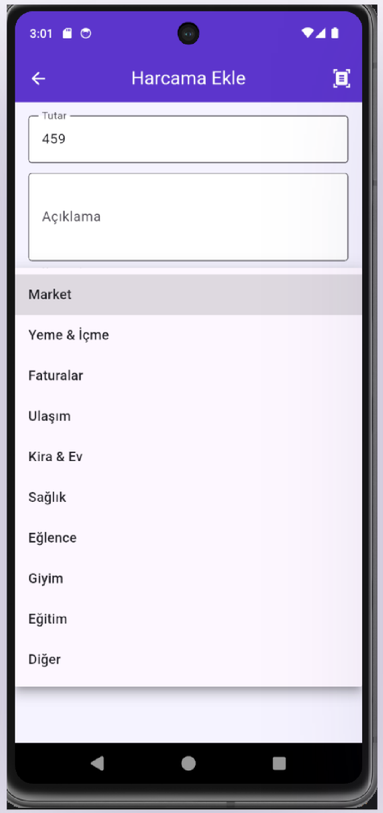
With data

MANUAL EXPENSE ENTRY & EDITING



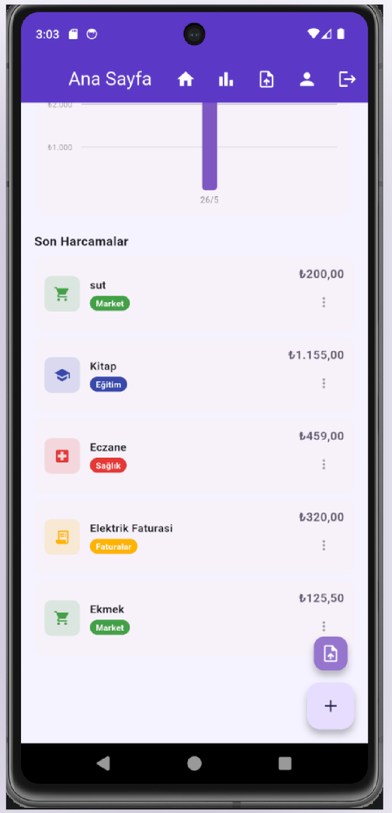
Add expense

Amount · description · category



Ten categories

Shared across every layer



Recent list

Edit & delete via context menu

STATISTICS — FOUR TABS



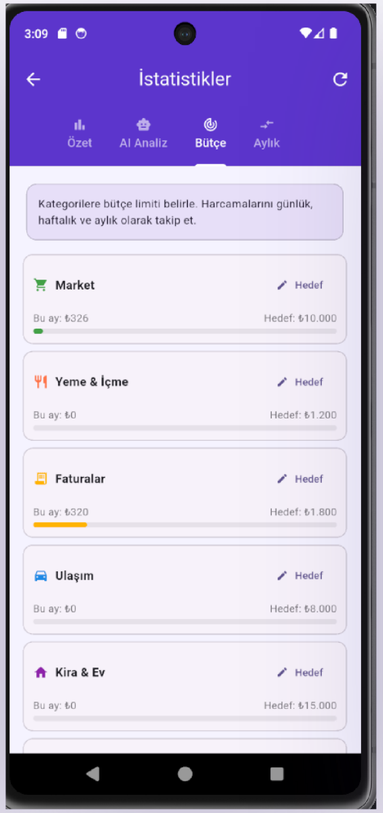
Özet

Totals & donut



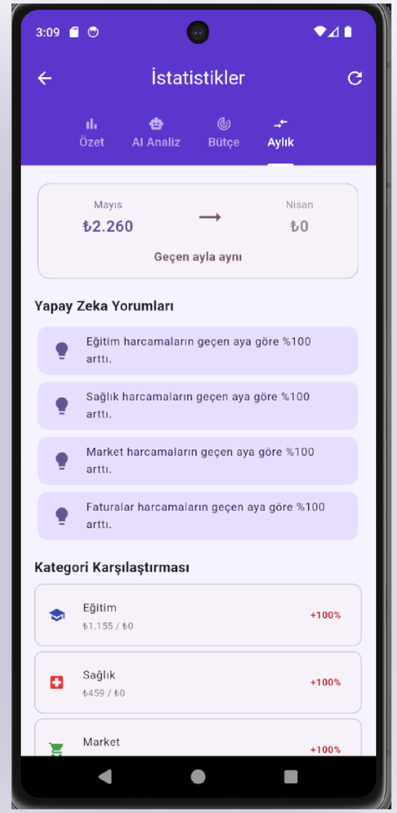
AI Analiz

Commentary & anomalies



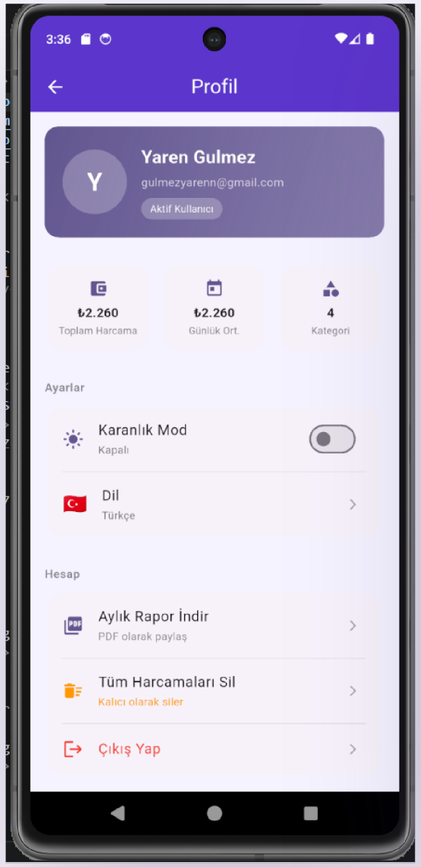
Bütçe

Per-category limits

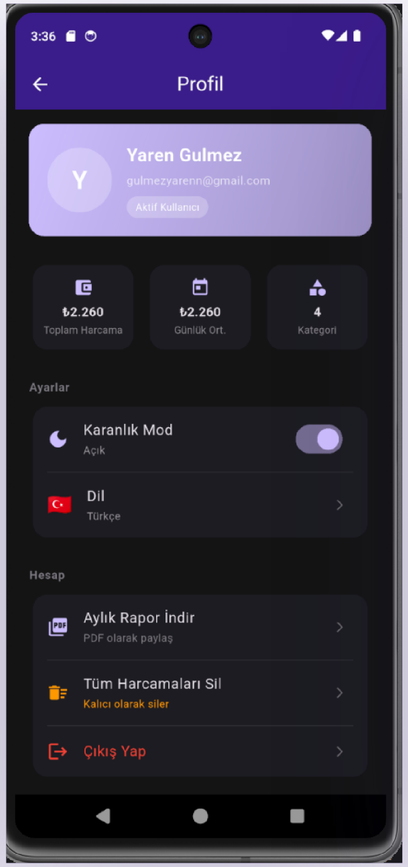


Aylık

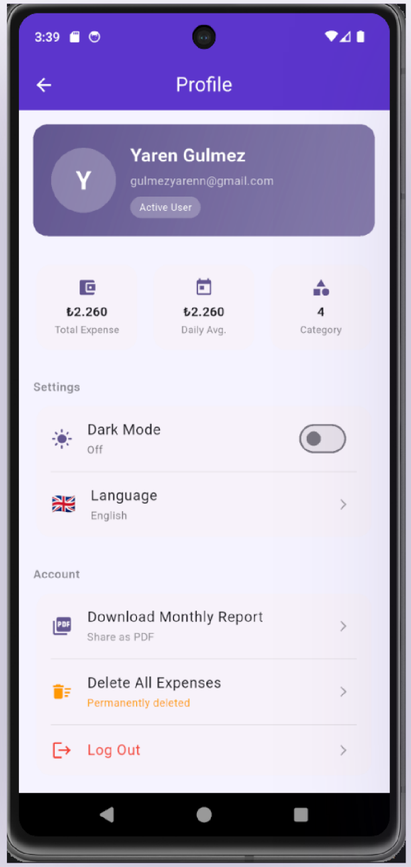
Month-over-month



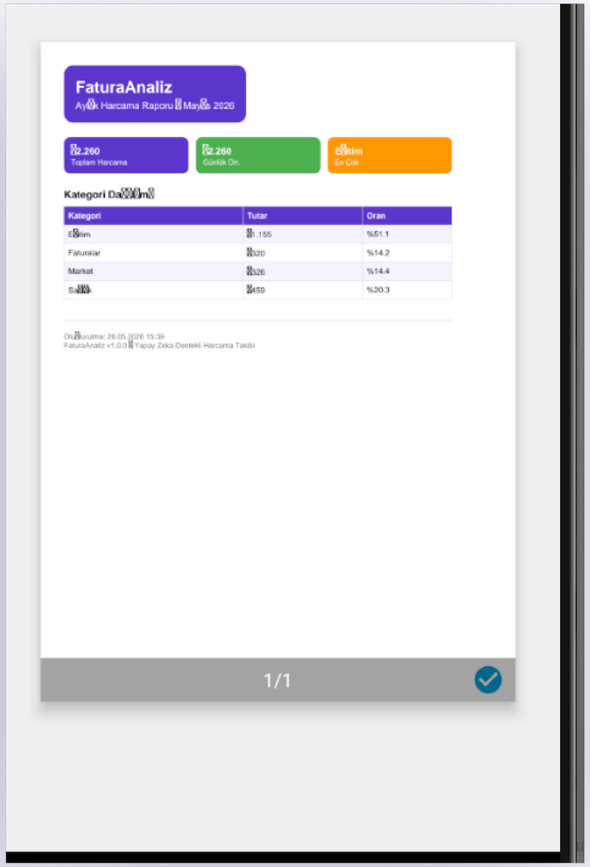
Profile
Light theme



Dark mode
Full re-theme



English
TR / EN switch



PDF report
Monthly summary

IMPLEMENTATION STACK

MOBILE

Flutter · Dart

http · Provider · fl_chart · image_picker · intl · pdf/printing

BACKEND

Spring Boot 3

Spring Security · Spring Data JPA · jjwt · JavaMailSender

AI SERVICE

Python · Flask

google-generativeai SDK · pandas · Gemini 2.5 Flash

DATABASE

PostgreSQL

JPA / Hibernate ORM · ACID · per-user isolation

PRESENTATION 03 — DESIGN & DEVELOPMENT

Thank you.
Questions & discussion.

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SUPERVISOR
Dr. Büşra Takgil

COURSE
CE498 · 2025–2026 Spring